

Soumyaratna Debnath

debnathsoumyaratna@iitgn.ac.in

Final Year Postgraduate (M.Tech)

Department of Computer Science and Engineering

Indian Institute of Technology (IIT) Gandhinagar

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Educational Qualification

DEGREE	INSTITUTION	CPI / CGPA	YEAR
M.Tech W&S (CSE)	IIT Gandhinagar, Gandhinagar, India	9.25 / 10	2022-2025
B.Tech (CSE)	CGEC, West Bengal, India	9.34 / 10	2018-2022
Class XII	APS Binnaguri, West Bengal, India	92.6 %	2018
Class X	APS Binnaguri, West Bengal, India	10 / 10	2016

Experiences

- Researcher at CVIG Lab, IIT Gandhinagar** **Mar 2024 – Present**
Serving as a *Project Fellow* at the Computer Vision, Imaging, and Graphics (CVIG) Lab, IIT Gandhinagar, under the mentorship of [Dr. Shanmuganathan Raman](#). I have contributed to the *Smart Farming* project with Larsen & Toubro Technology Services and the *Smart City Initiative* under the AI Centre of Excellence (AICoE), Ministry of Education, Government of India.
- Software Developer at IMS, IIT Gandhinagar** **Aug 2022 – Feb 2024**
Developed and maintained modules of the Institute Management System (IMS), an ERP solution for IIT Gandhinagar, to streamline operations across departments and facilities.
- Research Intern, Department of CSE, CGEC, West Bengal** **Sept 2020 – Feb 2022**
Conducted research on soft computing and meta-heuristics-based algorithms for multi-level image segmentation and thresholding. These works were concluded in two conference publications and a book chapter.

Publications

- Shadow Guided Optimal Packing and Assembly of General 3D Shapes** [Under peer-review](#)
Soumyaratna Debnath, Ashish Tiwari, Kaustubh Sadekar and Shanmuganathan Raman
- STEP: Simultaneous Tracking and Estimation of Pose for Animals and Humans** [Under peer-review](#)
Shashikant Verma, Harish Khatti, **Soumyaratna Debnath**, Yamuna Swami and Shanmuganathan Raman
- L3D-Pose: Lifting Pose for 3D Avatars from a Single Camera in the Wild**
[Accepted in International Conference on Acoustics, Speech, and Signal Processing \(ICASSP\) 2025](#)
Soumyaratna Debnath, Shashikant Verma, Harish Khatti, and Shanmuganathan Raman
- ScribGen: Generating Scribble Art through Metaheuristics**
[Published in SIGGRAPH Asia 2024](#)
Soumyaratna Debnath, Ashish Tiwari and Shanmuganathan Raman
- Modified Harris Hawk Optimization Algorithm for Multi-level Image Thresholding**
[Published as a Chapter in Hybrid Computational Intelligent Systems](#)
Soumyaratna Debnath, Abhirup Deb, Sourav De, Sandip Dey
- A New Modified Red Deer Algorithm for Multi-level Image Thresholding**
[2020 Fifth International Conference on Research in Computational Intelligence and Communication Networks](#)
Sourav De, Sandip Dey, **Soumyaratna Debnath**, Abhirup Deb

Research Projects

1. **Shadow Guided Optimal Packing and Assembly of General 3D Shapes.** Working on a novel approach for optimising the packing of arbitrarily shaped objects by analysing shadows cast from different angles and extending this technique to guide the precise assembly of object components.
2. **AI-powered Traffic Management Solutions for Sustainable Cities.** Worked on a deployable AI-powered traffic management solution, *IndAIINagar*, to monitor urban traffic using CCTV and drones in Indian settings. The project was undertaken by the AI Centre of Excellence (AICoE) under the Ministry of Education, Government of India.
3. **Analysis of Cotton Crop Fields using Aerial Imagery.** Leveraged drone-captured imagery to monitor and analyse cotton crops in agricultural fields. The initiative was carried out in collaboration with L&T Technology Services.
4. **Simultaneous Tracking and Estimation of Pose for Animals and Humans.** Conducted research to develop a novel transformer-based architecture for simultaneously tracking and estimating poses in primates for wildlife monitoring and behavioural studies. The research was in collaboration with [Dr. Harish Katti](#) from the Laboratory of Neuropsychology (LN), National Institute of Mental Health (NIMH), United States.
5. **L3D-Pose.** The research involved lifting 2D poses to 3D using an Attention-driven MLP architecture and accurately retargeting them onto 3D avatars; in collaboration with [Dr. Harish Katti](#), NIMH, United States.
6. **ScribGen.** The research involved designing meta-heuristics algorithms that could interpret and transform images into aesthetically pleasing scribble artworks.

Achievements

1. Finalist, [Qualcomm Innovation Fellowship India 2024-25](#) for the project proposal “*Shadow Guided Optimal Packing and Assembly of General 3D Shapes*”.
2. Qualified National Eligibility Test (NET) 2024 (96.43 percentile)
3. Qualified GATE 2021 (92.8 percentile) and GATE 2022 (96.11 percentile)
4. Granted a Copyright from the Government of India for the software project “*Picrypt It*”, which introduces an innovative method for embedding password-protected texts within colour images.
5. Finalist, *NanoArtography 2023*, an internationally acclaimed science image competition organised by Anasori Lab at Purdue University.
6. Awarded the Chairman Medal of Excellence for extraordinary academic performance from APS Binnaguri – 2018.
7. Awarded a Certificate of Merit from the Central Board of Secondary Education for outstanding academic performance (*a perfect 10 CGPA*) in the All India Secondary School Examination 2016.

References

1. **Dr. Shanmuganathan Raman**
Jibaben Patel Chair Professor, Computer Science & Engineering (Jointly with Electrical Engineering)
Indian Institute of Technology Gandhinagar, India
shanmuga@iitgn.ac.in
2. **Dr. Harish Katti**
Research Fellow, Laboratory of Neuropsychology (LN)
National Institute of Mental Health (NIMH), United States.
kattih2@nih.gov